Cattle Breed Detection and Categorization Using Image Processing and Machine Learning

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Abstract— **Researchers have had numerous opportunities to construct new analytical procedures in various scientific domains due to the development of machine learning and soft computing techniques. Livestock identification procedures are used to categorize different breeds of cattle based on their size and colour. A distinctive radio frequency, image frequency, or ear tag is necessary for the current methods used to predict specific breeds of cattle. We advise using different cattle picture datasets for the identification of the breed of cattle. The front, back, and sides of cattle will be photographed to extract key features, and these feature extractions will be carried out to uniquely identify each cattle with its information. The breed of cattle will be identified based on the processed images from each set of image data. Following picture preprocessing, each cow's images will be divided up by breed. Based on a preprocessed image dataset of cattle breeds, another category of rare cattle breeds is identified. In this study, the images of cattle facial morphology and various body shapes are recognized using the SVM machine learning algorithm. We may learn about Breed's fundamental characteristics, such as colour, average milk production, average body mass, etc., thanks to the proposed approach.**

Keywords— ***Image Processing, Image classification, Machine learning, Supervised learning, Support Vector Machine.***

I. Introduction

Several prediction applications and computer-assisted picture analysis are frequently utilized in a variety of human endeavors, including agriculture and animal husbandry. There are now numerous methods for determining the breeds of cows. One of the cutting-edge technologies that can be used to identify the breed is machine learning. Among the subcategories of supervised learning is the Support Vector Machine. Artificial neural networks and decision trees are two examples of machine learning approaches that are becoming more and more popular because of faster, stronger, and better adaptable tools used for classification and prediction, especially those related to nonlinear systems. The first step entails gathering all of the photos of the various cow breeds into a dataset. Next, the images are pre-processed by converting them into a greyscale image using Gaussian filtering which returns low-frequency values. The SVM model was trained using a real-time implementation model. Besides, a system-based Graphical User Interface was also developed.

In this study, a method for extracting features from processed photos using a support vector machine is proposed. Further, we explore the possibilities of hybrid intelligent systems that mix fuzzy logic and artificial neural networks. We also attempt to compute the breeding values of Holstein cows for milk and fat production based on their performance data and the EBV of their parents.

II. Literature Survey

In their article, Kumar, S. and Singh, S.K., explore how the characteristics of biometrics play a key role in how recent research in animal biometrics produces a qualified assessment of the general visual appearance of animals. Cow nose prints are used all over the world to track and identify the health of claiming livestock. There are certain guidelines for classifying cattle. The suitable cow body picture pattern is used to identify each cow using essential biometric characteristics [1].

The current demand for active cattle identification employing traceability for inquiry centres on Barry, B., et al works, which uses a sophisticated 3-step algorithm to accomplish nasal fingerprint scanning in all situations for muzzle fingerprint by lifting ink printing method. The main component analysis is used by the Euclidean distance classifier. The biometric recognition engine makes use of patterns that resemble horns, the body, and the nose. Each phase is completed using sophisticated ink. To extract patterns specific to these digital photos, the classifier employs cattle examples with group values [2].

To ensure correct breeding, Qiao, Y., et al. identify individual cattle. Ear tagging, radio frequency chips, and visual image processing are used for breed identification. We offer a deeper learning framework for recognising beef cow breeds from image sequences. The LSTM framework is used by advanced CNN to extract features from the front and back pictures of cattle. The enhanced version of V3 creates the image dataset with the network using extracted data. An algorithm for feature extraction is used throughout all procedures [3].

The biometric nose mark used to identify cows is discussed by El-Henawy, I., et al. The most recent technology,

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which resembles the sophisticated artificial neural networks employed by veterinarians, is used for bovine herds. The approaches employed here include pre-treatment methods such pre-treatment, advanced feature extraction, animal identification, histogram equalisation, box counting algorithms, mathematical morphological filtering, and segmentation-based fractal texture analysis [4].

According to Andrew, W., the process of visually distinguishing between similar items frequently necessitates multiple inspections. In visual bioassays, the recording of picture data is done using a single image method, an iterative process, and cutting-edge paradigm manipulation techniques. Individual photos, however, are insufficient, particularly for fine-grained object classifications. Several integrated paradigms with different benefits for animal identification are proposed by the animal biometric process to capitalise on human intuition for animal-based classification. The approach adopted is to deploy active robotic agents in agricultural-related situations to recognise individual biometric units and carry out online herd identification [5].

Using livestock ownership attribution for cow recognition and traceability of animal health analysis, coupled with vaccination and production control of important animals, Kumar, S. and Singh, S.K., presented new study fields in computer vision, pattern recognition, and cognitive science to comprehend the ailment. Individual farm animals are identified using conventional methods, such as electrical notches with ear tags, microchips with integrated tag freezes, and electrical tagging procedures. The techniques applied in this study are non-biometric strategies based on traditional techniques for animal detection, such as pattern recognition in traps [6].

Y. Qiao et al., Cattle are analysed using Enhanced Inception v3 and Advanced LSTM on a 30-frame video dataset. Cows' movements are monitored using radiofrequency ear tags. Advanced bidirectional long-term, and short-term storage techniques and advanced Inception v3 algorithms are employed in advanced convolutional neural network image sequences. Utilizing the BiLSTM extension layer, the cow rear view is used to extract data from the cow video dataset to extract time information that enables the cow to be uniquely identified. Cows can also be identified using photos taken from the side, back, and front [7].

Sian, C., et al. suggested a new nose print detection method that would use a projection algorithm to identify cows in real time using pictures of camera cows that experienced fishers had taken in the field. This technique makes use of wireless technology for server training. The database makes use of extraction for nose print recognition. Networked real-time security cameras employ each technique to recognise cows based on their nose prints. Given that each cow has a unique nose mark used to identify them, this form of detection is relatively challenging to accomplish. The mouth pattern analysis method for protocol benchmarking is equipped with a server that archives acquired photos, extracts key characteristics of the detection approach, and performs evaluation and comparison analysis using different algorithms [8].

S. Kumar et al., Bovine pattern prints served as the foundation for the computer vision employed for bovine biometrics. for computerised methods of animal identification and registration. The features are extracted using the 'cow nose print' of the applied learning network, and automatic coding is then applied to the diagnostic table stack of extracted coding and decoding features. A new strategy to identify cattle is to extract features using cattle texture analyses for representation using deep learning networks. This method can be used for noise reduction, stack auto coding, or auto coding of any method. An approach known as stacking coding uses each component of the nasal print analysis [9].

The moving herd of individual Holstein cows is recognised by a body covering pattern, and Bello, R.W., et al. incorporate deep learning with computationally improved UAV Advanced M100. In this paper, a method using three deep learnings is described that makes use of sophisticated classical neural networks. Inception v3's advanced biometric base and exploratory deployment agency two CNN data streams are combined to identify specific Friesian cattle, while YOLO v2 detectors are utilised to determine species [10].

III. PROPOSED SYSTEM

The proposed system offers a vigorous and hardy approach to the selection of unique livestock and morphologically distinct cattle breeds. Fig 1 depicts the architecture of the proposed system.

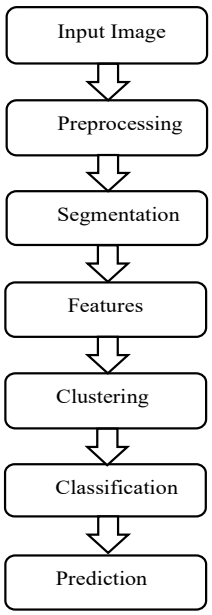

Fig.1. The architecture of the Proposed System.

A. DATASET

In this model, over 1000 images were collected as a dataset. The dataset was categorized into seven types of cattle in different regions of India and other countries. Images like Fig.3 were fed to the dataset.

The model recognizes seven varieties of cattle breeds:

1. Ayrshire
2. Brown Swiss
3. Holstein Friesian
4. Jersey
5. Kankrej
6. Malvi
7. Red Dane

The Fig.2. shows the composition of these seven cow breeds in the dataset.

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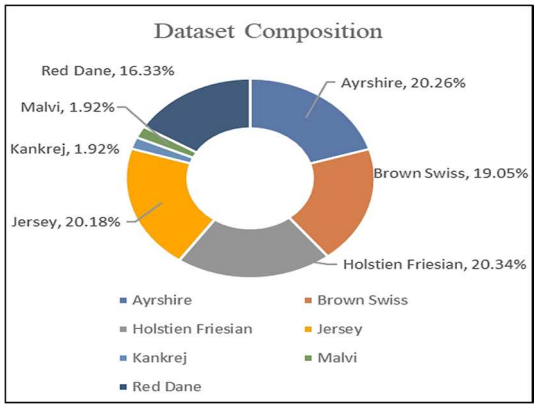

Fig.2. Dataset Composition.

B. PRE-PROCESSING

This step of pre-processing the data involves the steps needed to transform or encode the data such that it can be easily analyzed on the machine**.** For a predictive model to be accurate, the main goal of the algorithm is to interpret the characteristics of the data easily. In this step, the image was pre-processed after it was loaded into the GUI. Pre-processing removes noise from the image.

C. IMAGE PROCESSING

The method for performing certain operations on an image constitutes image processing and is undertaken to obtain an improved image and/or extract useful attributes or information from it. This is a type of signal processing where the input may be an image and the output may be an enhanced image or a property or function linked to that image.

D. GREY SCALING

Grey scaling is the process of transforming an image from another colour space e.g., RGB, CMYK, HSV, etc. It differs between all black and all white. In digital images, greyscale means that the value of each pixel represents only light-intensity information. Such images usually only display from the darkest black to the lightest white. That is, the image contains only black, white, and grey and has multiple levels**.** The Fig.4. shows the greyscale image that's processed from the original raw image. The critical steps involved in using the greyscale algorithm are:

- Step 1: Extract and Obtain the values for the pixel in the form of red, green, and blue.
- Step 2: After the extraction utilizes computational mathematical functions to metamorphose these RGB figures into an exclusive grey number.
- Step 3: Restore the indigenous red, green and blue values in the image with the exclusive grey numbers.

Weighted Greyscale Method:
In the weighting method, also known as the luminance method, red, green, and blue are weighted according to the wavelength. Following is the formula for calculating the weighted greyscale value:

$$G = 0.299R + 0.586G + 0.114B \qquad (1)$$

Here G is the greyscale value, and intensity values of red, green, and blue are expressed as R, G, and B respectively.

WATERSHED

Watershed algorithms are used in image processing mainly for object segmentation, i.e., to separate different objects in an image. This allows counted objects or separate objects to be analyzed further. The watershed algorithm is classically used for segmentation, particularly used in extracting objects that touch or overlap in an image. There are different technical definitions of watersheds e.g. In a graph, watersheds can be demarcated on nodes, on edges, as hybrids on nodes as well as edges or in the continuum. Various algorithms are used for calculating watersheds. Source classification algorithms are used in image processing mainly for object segmentation, i.e., to separate different objects in an image. This allows counted objects or separate objects to be analyzed further. The Fig.5. shows the watershed image.

The steps for the greyscale algorithm are:

• Step 1:	Read the input colour image and transform it to a greyscale image.
• Step 2:	Utilize the Gradient Magnitude as the Segmentation Function for the greyscale image.
• Step 3:	Highlight the foreground object in the image.
• Step 4:	After highlighting the foreground object Compute Background Markers.
• Step 5:	The next step involves to Compute the Watershed Transformation from the Segmentation Function.
• Step 6:	Conceive the outcomes.

Fig.3. Original Image.

Fig.4. Greyscale Image.

Fig.5. Watershed Image.

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SEGMENTATION

Grouping pixels together to represent compact performance constitutes segmentation. In the area, it is based on segmentation to directly define a particular area. The method is employed in an image to obtain pixel segmentation values [21]. It is based on the seed point (early) of the image after the cow images have been converted to greyscale, and image classification into a certain number of classes is done. The foreground and background of the cow image are segmented separately at each stage. In the last step, the cow's body parts are individually segmented.

E. FEATURE EXTRACTION

It is a dimensionality reduction process wherein initially a set of raw data is reduced into a more manageable group for processing [22-24]. The hallmark of such a bulky database is the huge number of variables that consume vital computational resources during processing. Feature extraction thus helps by selecting and/or combining related variables thereby effectively reducing the amount of data to be processed without compromising the attributes or accuracy of the initial dataset.

The scale-invariant feature transform algorithm was used to extract each of the discreet local morphological features of the cattle like horns, ears, back humps, coat patterns, etc.

F. CLASSIFICATION

In classification, the program learns from the specified training dataset (supervised classification) or observation and classifies the new observation into a set of classes or groups.

We have collected over 1000 images and trained the SVM Classifier model to compute raw cattle images. Each step is performed on individual cattle images and based on this, the breeds are classified and recognized as shown in Fig.6 & Fig.7.

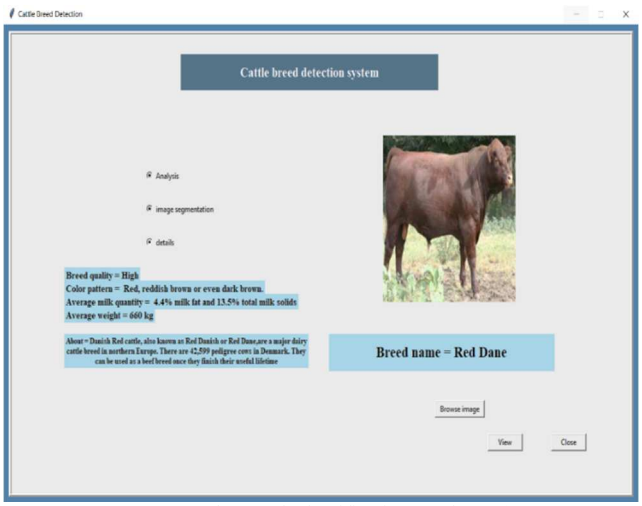

Fig.6. Cattle Breed Classification (Red Dane).

SUPPORT VECTOR MACHINE

SVM is a supervised machine learning algorithm used for classification or regression predicaments. It transforms the data using a technique called kernel tricks and

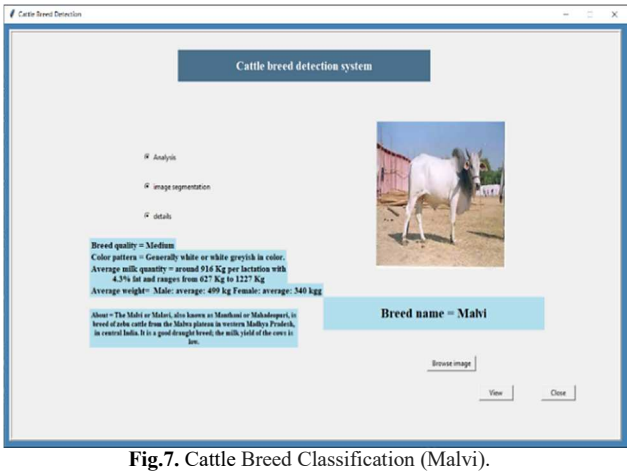

Fig.7. Cattle Breed Classification (Malvi).

finds the optimal boundaries between possible outputs based on those transformations. The goal of the SVM algorithm is to create optimal lines or decision boundaries that can divide n-dimensional space into classes so that new data points can be easily placed in the correct category in the future. This best decision boundary is called the hyperplane.

The SVM algorithm steps include the following:

• Step 1:	Load the essential data science libraries.
• Step 2:	Import the dataset and take out the X1 and X2 variables separately.
• Step 3:	Split the image dataset into training and testing subsets.
• Step 4:	Commence the SVM categorizing model.
• Step 5:	Fixing and fitting the SVM classifier model.
• Step 6:	Predict the Class.

Following is the formula for the polynomial kernel:

$$(X1, X2) = (X1^T . X2 + 1)^d \qquad (2)$$

Here *X1* and *X2* feature, *d* is the degree of the polynomial, which we need to specify manually.

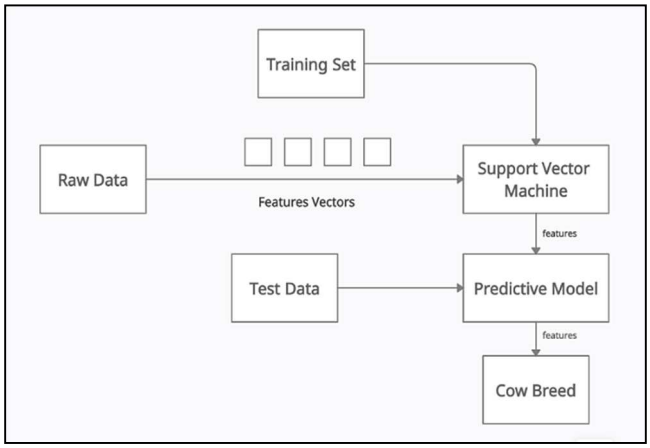

Fig.8. Predictive Model Architecture

Predictive modelling is modelling a concept for creating models that can make predictions. Such models typically include machine learning algorithms that learn specific properties from training datasets and make predictions about them.

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IV. RESULT

A. MODEL TESTING AND ACCURACY

A total of 376 instances of queries were tested on the model, the Table-I shows us the result of these testing queries on the classifier model.

TABLE I
TEST MATRIX

		True Class							Total
		A	BS	HF	J	K	M	RD	
Predicted Class	A	50	1	1	0	0	0	2	54
	BS	1	49	0	2	0	1	1	54
	HF	0	2	48	0	1	1	0	52
	J	1	0	0	49	3	1	0	54
	K	1	1	0	2	50	0	0	54
	M	1	0	2	0	0	51	0	54
	RD	2	0	2	0	1	1	48	54

Abbreviations:

- Ayrshire (A)
- Brown Swiss (BS)
- Holstein Friesian (HF)
- Jersey (J)
- Kankrej (K)
- Malvi (M)
- Red Dane (RD)

Following is the formula to calculate the accuracy of a class by Equation (3):

$$A_c = \frac{Number\ of\ Positive\ Results \times 100}{Total\ number\ of\ predictions} \quad (3)$$

Here A_c stands for Class Accuracy given by Equation (4)

$$A_M = \frac{\sum_1^n A_c}{n} \quad (4)$$

Here A_M stands for Model Accuracy, A_c stands for Class Accuracy in percentage and n stands for the Number of Classes.

TABLE II
MODEL ACCURACY

Class	Positive Results	Total	Class Accuracy (%)	Model Accuracy (%)
Ayrshire	50	54	92.59	91.76
Brown Swiss	49	54	90.74	
Holstein Friesian	48	52	92.31	
Jersey	49	54	90.74	
Kankrej	50	54	92.59	
Malvi	51	54	94.44	
Red Dane	48	54	88.89	

Accuracy graph Fig.9 plotted with the help of Table II.

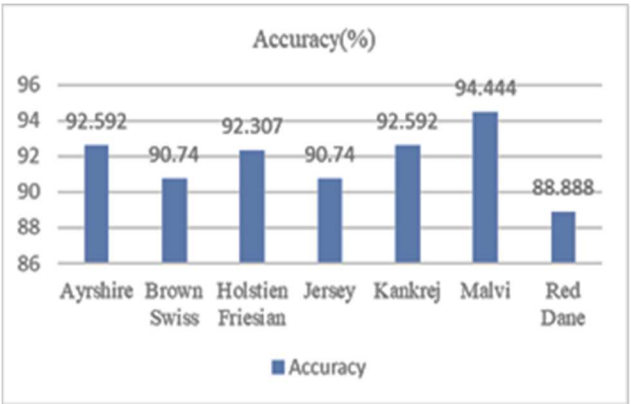

Fig.9. Accuracy Graph for Each Breed Classification

IV. CONCLUSIONS AND WAY FORWARD

The dairy sector is one of the largest agricultural industries in the world and a key economic force in every nation. Since there are fewer native breeds of cattle and more crossbreeds, automatic identification of cow breeds is crucial. More than 100 cow image datasets were gathered for our suggested system, and these photos underwent a pre-processing step where the noise was removed, and a certain dimension was changed. Then, the various cattle body sections were retrieved using the SIFT approach for feature extraction. Finally, we used SVM to divide the cattle into 7 classes and forecast the breed. This suggested technique only functions provided the cattle image is a comprehensive representation of a single animal. Future studies will explore identifying cattle in photos with other animals present but against other backdrops. Cattle have been further separated, with many varieties of cattle depicted. We may learn about Breed's fundamental characteristics, such as colour, average milk production, average body mass, etc., thanks to the proposed approach.

To adequately train the algorithm, future work will need a larger collection of cow photos from each viewpoint variation. The current model may experiment with many techniques, like convolutional neural networks (CNN). The model can eventually incorporate information on additional cows, additional types of cows, and other animals as necessary. The entire model may get slightly heavier by including more cow breeds and training the model with more photos taken from more angles. A large dataset will be required for the model. For better outcomes, developers will need to shoot their photographs as opposed to depending on available datasets. Future applications will also include a camera feature that, when launched, quickly scans the cow to reveal its breed without the user having to click on the image.

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